

Sri Lanka Institute of Information Technology



Lab Report
Lab sheet No 01

2026-Y2-S1-KU-01

Web-based Vehicle service and fuel station management System

Group Members

Name	IT Number
IT23642096	MANATHUNGE A.I
IT24104390	Ahamed M.N.A
IT25100002	Sarma S S L
IT25100020	Nayanajith K.A.T
IT25100117	Herath H.M.A.R.B
IT25100119	Liyanage D.D.S

AutoFlow Sri Lanka

Requirement Gathering Report

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MODULE

SE2030

Software Engineering

PROJECT TOPIC

Vehicle Service & Fuel Management System

GROUP

[2026-Y2-S1-KU-01]

GROUP MEMBERS & CORE OWNERSHIP

01

IT23642096|MANATHUNGE A.I —
Vehicle Management

02

IT24104390|Ahamed M.N.A
— Service
Booking Management

03

[IT25100002 | Sarma S S L—
Fuel Management

04

[IT25100020 | Nayanajith K.A.T—
Payment Management

05

[IT25100117 |Herath H.M.A.R.B]
— Notification
Management

06

[IT25100119 | Liyanage D.D.S]
— User Management

01 Document Overview

Purpose, business value, project details and document control

Contents

01	Document Overview	02
02	Requirement Details	03
03	As-Is / To-Be / BPM	04
04	Core CRUD Allocation	05
05	Core Details: Members 1-3	06
06	Core Details: Members 4-6	07
07	Rules, UX, Reports & Quality	08
08	Evidence Samples 1-3	09
09	Evidence Samples 4-6	10
A-E	Complete Conversation Detail & Conclusion	11-15

PURPOSE OF URF

Turn stakeholder needs into a clear build plan.

This form records the users, workflows, six core CRUD areas, business rules, reports, quality needs and decisions that still need approval.

REQUIREMENT DETAILS FOR PROPOSED PROJECT

Project	Vehicle Service and Fuel Management System
Scope	Vehicle records, service booking, fuel tracking, payment, notifications, users and station operations.
Method	Project Discussion Bot personas, normal flows, exceptions and gap checks.
Requested	24 July 2026
Expected	23 September 2026
Status	Complete Draft 1 — team review required

1

One trusted source

Vehicles, bookings, fuel logs, payments and customer records stay connected.

2

Faster customer service

Quick search, simple booking, verified payment and a clear digital invoice.

3

Fewer mistakes

Reduce repeat entry, bay conflicts and unclear change ownership.

4

Better breakdown control

Find affected customers and send clear alternatives or refunds.

5

Strong financial control

Verification, reconciliation, refund review and audit history.

6

Useful planning data

Demand, occupancy, fuel-cost trends and performance reports.

Evidence rule — Pages 9-10 must contain six genuine Project Discussion Bot screenshots. Appendices A-E preserve the detailed meeting findings.

VERSION	DATE	CHANGE	PREPARED BY	REVIEW / APPROVAL
1	24/07/2026	Requirements grouped		Team review

02 Requirement Details

Stakeholders, role access, information and requirement status

SCOPE STATEMENT

Connect vehicle records, customer booking, fuel tracking, payments, notifications and user access. Support refunds, complaints, maintenance history, fuel discrepancies and breakdown incidents.

CUSTOMER-SIDE ROLES

Public visitor

Open access

Services, fuel plans, prices, station locations, delay/breakdown notices, contact details, FAQs and terms.

Guest customer

Quick booking

Book service without an account. Secure invoice and refund retrieval method is open.

Registered customer

Own records

Bookings, vehicles, invoices, fuel logs, service history, refunds, complaints and personalised alerts.

STAFF AND DECISION ROLES

Dilmi Jayasinghe

Customer Service

Booking support, complaints, refund requests and customer messages.

Ruwan Fernando

Scheduling

Bookings, bay/technician allocation and conflict checks; recommends cancellation.

Malsha Wickramaratne

Finance

Payments, reconciliation, duplicates and authorised refund approval.

Niroshan Perera

Operations: breakdowns, operational changes, incidents and reports.

Kasun Senanayake

Fuel Station Manager: dispensing, bay changes, vehicle information, escalation, incidents and daily station reports.

Admin / Management

Accounts, roles, audit and dashboards. Sensitive limits and cancellation boundary are open.

Information to Capture

- Account - proposed:** name, NIC/license no., DOB, mobile, email, username and protected password.
- Vehicle:** registration no., make/model, fuel type, engine capacity, owner and last service/mileage.
- Booking:** contact, vehicle, service type, center, date/time and technician preference.
- Payment:** booking/transaction IDs, method, amount, date/time, payment and refund status.
- Fuel log:** vehicle, station, fuel type, quantity, odometer reading, amount and date/time.
- Complaint / incident:** IDs, category, description, evidence, related vehicle/booking and status.

Requirement Status

CONFIRMED

 Agreed and safe to design.

PROPOSED

 Useful value; approval needed.

OPEN

 Gap or conflict still unanswered.

Privacy rule: customers see only their own records. Staff access follows role and need. Scheduling normally receives aggregate demand/bay data, not payment, complaint or fuel-billing data. Important edits record user, action, date/time, old value and new value.

REQUIREMENT GATHERING SUMMARY

The meeting covered normal service journeys, payment/fuel failure, refunds, breakdown handling, bay/technician conflict, fuel discrepancy, permissions, reporting, security, backup, record retention and future ideas. Wider groups: customers/public, workshop operations, management, technical/security, fuel suppliers and regulators.

How the fragmented process becomes one connected customer service

Duplicate entry, slow inter-station coordination, isolated fuel records and time-consuming daily report consolidation.

One shared platform with current bay status, technician allocation, fuel stock and customer alerts.

Guest / sign in Select vehicle & service Choose center & slot Confirm booking Verified payment Digital invoice Service history & fuel log

Vehicle + technician/bay check → draft → conflict check → approve → publish → notify.

Success but no invoice → incident → retry → staff review → invoice or refund.

Detect meter/log mismatch → investigate → adjust/refund → notify → audit.

Request → eligibility → approval → payment → notify → close.

Register → assign → investigate → resolve → notify → close.

All bays full → join queue → bay released → notify customer.

Reserve one active bay slot or offer queue/another center.

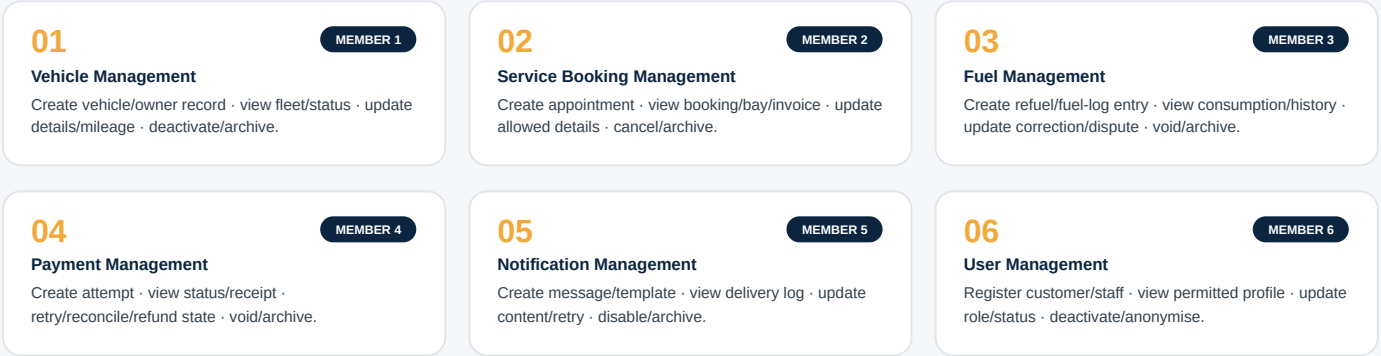
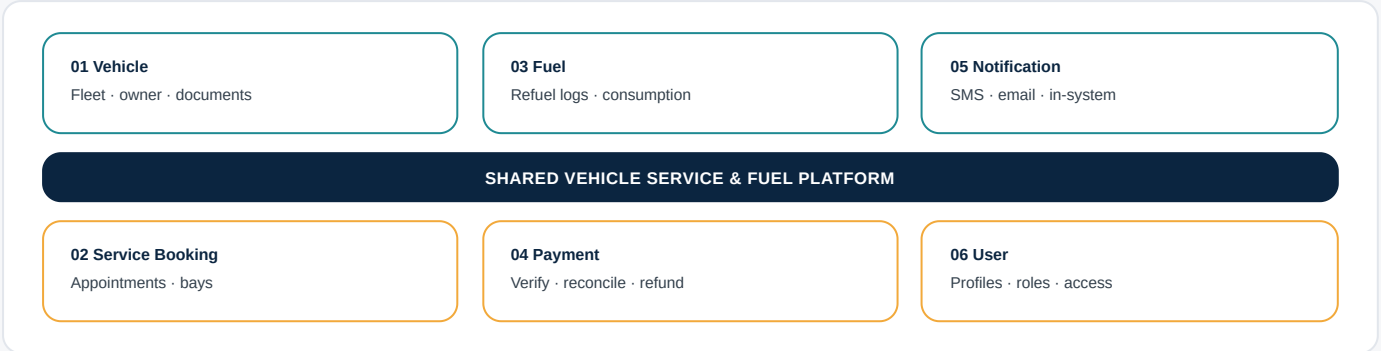
Invoice only after verification; keep pending attempts recoverable.

System flags for review; authorised role approves major action.

04

Six Core Functions

Exactly one major CRUD area for each of the six team members



Safe lifecycle — For operational and financial records, "Delete" means cancel, void, deactivate, archive or anonymise according to permission and retention rules. Keep the audit trail.

SHARED BOUNDARY	RULE
Bay & schedule	Booking reads live bay/technician status; changes identify affected appointments.
Payment & invoice	Invoice follows verified payment; both records stay linked and recoverable.
User & permission	Every staff action uses role rules and records user, time, old value and new value.

05 Core Details — Members 1 to 3

CRUD actions, sub-functions, rules and exception handling

01 Vehicle Management

MEMBER 1

Keep an accurate fleet record and stop unsafe or unregistered vehicles from being booked.

CREATE

Vehicle & owner records.

READ

Fleet, model, fuel type, status.

UPDATE

Details, mileage, ownership.

LIFECYCLE

Deactivate/archive; keep history.

- Registration no., make/model, fuel type, engine capacity and owner.
- Active, in-service and archived states.
- Missing document or insurance blocks booking.
- Ownership change triggers record transfer.
- Search/filter by plate, model, fuel type, status and date.

Rule / exception A vehicle with expired insurance/revision cannot be booked. Changes must identify affected bookings and reminders.

02 Service Booking Management

MEMBER 2

Reserve valid appointments and keep every booking, bay and invoice recoverable.

CREATE

One-time/recurring; guest checkout proposed.

READ

Booking ID, vehicle, bay, invoice.

UPDATE

Slot, technician, partial change.

LIFECYCLE

Cancel/archive with reason and audit.

- Center/date search, service type, bay and technician assignment.
- One bay-slot can have only one active booking.
- Invoice issue/reissue, history and lost-invoice recovery.
- Queue when full; offer released slots.
- Bay/technician/center changes: reassign, alternative or refund.

OPEN queue release, guest retrieval, fees/no-show and final bay hold.

Confirmed Payment before invoice · partial reschedule subject to bay availability · cancelled booking cannot be reused.

03 Fuel Management

MEMBER 3

Record accurate refuelling and consumption data and flag unusual fuel activity.

CREATE

Refuel / fuel-log entry.

READ

Vehicle, station, quantity, history.

UPDATE

Correction, dispute status.

LIFECYCLE

Void/archive disputed entry; keep history.

- Vehicle, station, fuel type, quantity, amount and odometer reading.
- Detect abnormal consumption or duplicate entries.
- Draft → verify → confirm → post-to-account workflow.
- Link fuel spend to vehicle running-cost reports.
- Search/filter by vehicle, station, date range and fuel type.

Rule / exception An entry with an odometer reading lower than the last record is flagged for review. Audit old/new values and reason.

06 Core Details — Members 4 to 6

Money, communication, identity and supporting CRUD areas

04 Payment Management

MEMBER 4

Verify every service and fuel payment, prevent duplicate loss and keep finance records ready for review.

CREATE

Attempt and linked transaction.

READ

Status, amount, receipt, response.

UPDATE

Retry, pending, verified, refund.

LIFECYCLE

Void/archive failed record; retain history.

- Verification before invoice; duplicate ID flag.
- Late gateway response after bay hold.
- Success + invoice failure: incident, retry, review.
- Gateway/bank reconciliation and reviewed refunds.

OPEN

exact providers, payment confirmed after released bay, final refund rates and automatic/manual amount limits.

05 Notification Management

MEMBER 5

Send timely information and show whether each message was delivered.

CREATE

Template, notice, event message.

READ

Templates, preference, delivery log.

UPDATE

Content, channel, retry, state.

LIFECYCLE

Disable/archive; keep the log.

- SMS, email and in-system messages.
- Booking, payment, reminder and service-due triggers.
- Delay, cancellation, bay/technician changes.
- Refund, complaint, low-fuel-balance, emergency/security.

OPEN

final timing, retry count, channel preference and emergency wording.

06 User Management

MEMBER 6

Give customers simple access and staff controlled access without exposing private data.

CREATE

Customer registration, staff account.

READ

Own profile or role-permitted data.

UPDATE

Profile, password, role, status.

LIFECYCLE

Deactivate/anonymise by policy.

- Register, login, OTP password recovery and profile.
- Guest checkout; retrieval method to confirm.
- RBAC for admin, operations, scheduling, finance, service and station.
- Audit changes and protect identity/payment data.

OPEN

exact administrator limits and final deletion/anonymisation.

ADDITIONAL CRUDS AND FUTURE ENHANCEMENTS — NOT EXTRA MEMBER ALLOCATIONS

Refund

Complaint

Emergency / Breakdown

Maintenance

Bay / Technician

Fuel Card

Route / Station

Waiting List

Group Booking

Feedback

Lost & Found

Document

Staff / Permission

Audit / Investigation

Pricing / Fare

Special Services

Employee Metrics

Loyalty

Advanced Analytics

07 Rules, UX, Reports and Quality

Confirmed controls, stakeholder proposals and decisions before design freeze

Business Rules

VALIDATED BASELINE

- One bay slot = one active booking.
- Verified payment before invoice; cancelled booking cannot be reused.
- Customer sees own data; staff use role access.
- Authorised operations cancels; finance approves refunds.
- Critical changes notify affected customers.
- Audit user, action, date/time and old/new values.
- Detect duplicate transaction IDs; send them to Finance review.

UI / UX

- Mobile-first, simple words, clear steps and status.
- Fast center/date search, filters, bay availability and price summary.
- Readable digital invoice and guest/registered dashboards.
- Staff work queues, warnings, approvals and history.
- Keyboard, screen reader, colour-safe, special assistance.
- English, Sinhala and Tamil; clear retry/recovery pages.

Reports & Search

- Revenue, invoices, refunds and reconciliation.
- Delay, cancellation, on-time, reason/duration and old/new bay change.
- Station/bay, complaints, staff and incidents.
- Demand, occupancy, peak-hour, fuel consumption and change impact.
- Search invoice/vehicle/customer/NIC/booking/date/center.
- PDF/Excel/CSV export is proposed; report access is role-restricted.

Integrations & Continuity

- Payment gateway and bank; exact provider/API open.
- SMS and email with retry and staff fallback.
- Existing workshop and fuel-station systems.
- Proposed encrypted offline invoice rules, then later sync.
- Daily backup, transaction logs and tested recovery are proposed.
- Gateway, network and single-station outages need final procedures.

≤ 3 sec

Service search proposal

≤ 10 sec

Booking target after payment

99.5%

Service uptime proposal

RTO 4 h

Restore critical service

RPO 15 m

Maximum data-loss window

STAKEHOLDER PROPOSALS

Bay hold: 15 minutes. Refund: booking cancelled 100%; >48h 90%; 24-48h 75%; <24h 50%; after service starts 0%. Fuel-discrepancy amount/threshold TBC; complete in 3-7 working days; small-auto/large-finance limit TBC.

Notice retention: delay to service end; cancellation 30d; refund completion +30d; emergency until cleared. Offline invoice: protected local list, then sync.

RETENTION PROPOSAL

Booking 5 years · payment/refund 7 years · complaint 3 years · incident 10 years · audit 5 years. Protected credentials, OTP, safe sessions, role access and audit are confirmed security needs. Final deletion/anonymisation is open.

OPEN DECISIONS BEFORE DESIGN FREEZE

OPEN

Multi-vehicle bookings: first release or future?

OPEN

Final refund, discrepancy eligibility and approval limits?

OPEN

Late payment after the bay hold expires?

OPEN

Guest invoice/refund/complaint retrieval and guest vs registration-first?

OPEN

Offline duplicate invoice action before synchronisation?

OPEN

Exact field-level permissions, Admin boundaries and major-incident escalation?

OPEN

Peak users/bookings and final browser/device/accessibility target?

OPEN

Deletion/anonymisation; permanent confirmation vs 5-year booking record?

OPEN

Exact payment, SMS, email and fuel-supplier vendors/APIs?

OPEN

Group/no-show and final notification timing/retry/preferences?

Page 8 of 15

08 Requirement Evidence — Set 1

Replace each frame with a genuine, readable Project Discussion Bot screenshot

SCREENSHOT STANDARD Show persona, student question and complete bot response. Keep readable at 100% PDF zoom.

SAMPLE 01

Dilmi / Customer Service · booking, cancellation, refund, complaint and customer data

thank you for your time. this is Dilum. Dilmi I want you to go through the tasks that you do daily, and how you do it.

 Project Discussion Bot - Year2

Dilmi Jayasinghe – Customer Service Executive

Hello Dilum. Thank you for taking the time to speak with me.

My day usually starts by checking new customer inquiries, service requests, and notifications received overnight. I log into the system and review any pending tasks that need attention.

When a customer wants to register a vehicle, I collect details such as the vehicle number, owner's name, contact details, vehicle type, and current mileage. I enter these details into the system and verify that there are no duplicate records.

A large part of my day is handling service bookings. Customers call, send messages, or visit our office to book a service. I check available dates, create the booking, and provide confirmation to

explain how you do each thing that you mentioned right now?

 Project Discussion Bot - Year2

Dilmi Jayasinghe – Customer Service Executive

Hello Dilum. Sure, I will explain each task step by step as I do it in my daily work.

Registering a New Vehicle

When a customer comes to our office or calls us, I first collect their details. I ask for the owner's name, phone number, address, vehicle registration number, vehicle type, make and model. I also ask for the current mileage.

Then I search the system to check whether the vehicle already exists. If it does not exist, I create a new vehicle record and save all the information. Finally, I verify the details with the customer before saving.

Updating Vehicle Details

SAMPLE 02

Niroshan / Operations · access, authority, breakdowns and business rules

Niroshan, This is Anjana. Please let me know how you do all the tasks now?

 Project Discussion Bot - Year2

Niroshan Perera – Workshop Operations Manager

Hello Anjana. Sure, I will explain exactly how I do my daily tasks.

When I start work, I open the system dashboard and check all the service bookings scheduled for the day. I look at the vehicle details, service type, booking time, and workshop bay availability. This helps me understand the workload for the day.

When a vehicle arrives, I search for the booking in the system and confirm the vehicle registration number and customer details. If everything is correct, I mark the vehicle as arrived and assign it to an available service bay.

consider all the facts that we have discussed so far and suggest anything that we may have missed.

 Project Discussion Bot - Year2

Niroshan Perera – Workshop Operations Manager

Hello Anjana. After our discussion, I think you have covered the main functions well. However, there are a few things that may have been missed during requirement gathering.

For **Vehicle Management**, we should keep vehicle documents such as insurance, revenue licence, and registration expiry dates. The system should remind users before these documents expire.

For **Service Booking Management**, it would be useful to record the service type, service priority (normal or urgent), technician assigned, and estimated completion date.

SAMPLE 03

Ruwan / Scheduling · bay/technician allocation, demand and change impact

ruwan im Anjali, do you also handle customer data for scheduling decisions?

 Project Discussion Bot - Year2

Ruwan Fernando – Service Scheduling Officer

Hello Anjali. Yes, I do use some customer data when making scheduling decisions, but only the information needed for booking and service operations.

When I schedule a service, I check the customer's vehicle details, previous service history, preferred appointment times, and any special requests they have made. This helps me assign the correct service type and estimate how much workshop time is needed.

For example, if a customer's vehicle had a major repair recently, I may allocate a longer inspection slot. If they regularly request morning appointments, I try to accommodate that if space is available.

ruwan this is Anjali. explain what are the difficulties you are facing

 Project Discussion Bot - Year2

Ruwan Fernando – Service Scheduling Officer

Hello Anjali. Yes, there are several difficulties I face in my daily work.

One common problem is **last-minute cancellations and rescheduling requests**. Sometimes customers cancel on the same day, which leaves workshop time slots unused. On the other hand, some customers want urgent appointments when the schedule is already full.

Another challenge is **limited workshop bay availability**. There are times when many vehicles need servicing at once, but we do not have enough bays or technicians available. This makes scheduling more difficult.

I also face issues when **vehicle information or customer details are incomplete or outdated**. If

09

Requirement Evidence — Set 2

Final evidence page and submission quality check

SCREENSHOT STANDARD Show persona, student question and complete bot response. Keep readable at 100% PDF zoom.

SAMPLE 04

Malsha / Finance · revenue, payment, reconciliation, refund and approval

thank you for your time. this is Lojan. Malsha I want you to go through the tasks that you do daily, and how you do it.

Project Discussion Bot - Year2

Malsha Wickramaratne – Finance & Revenue Manager

Hello Lojan. Thank you for speaking with me.

My main responsibility is managing payments, revenue records, and financial tracking. Every day, I use the system to monitor customer payments and ensure all financial transactions are recorded correctly.

When a customer makes a payment, I first search for the related invoice. I verify the amount due and the payment method used, such as cash, card, or bank transfer. Then I confirm the payment and update its status in the system.

thank you for your time. this is Lojan. Malsha I want you to go through the tasks that you do daily, and how you do it.

Project Discussion Bot - Year2

Malsha Wickramaratne – Finance & Revenue Manager

Hello Lojan. Thank you for speaking with me.

My main responsibility is managing payments, revenue records, and financial tracking. Every day, I use the system to monitor customer payments and ensure all financial transactions are recorded correctly.

When a customer makes a payment, I first search for the related invoice. I verify the amount due and the payment method used, such as cash, card, or bank transfer. Then I confirm the payment and update its status in the system.

SAMPLE 05

Kasun / Fuel Station · daily work, dispensing, incidents and live dashboard

Kasun, I am Amantha. please go through everything that you do daily and how you do them

Project Discussion Bot - Year2

Kasun Senanayake – Fuel Station Manager (Colombo Depot)

Hello Amantha. I'd be happy to explain my daily work and how I do each task.

My day starts by checking the fuel inventory levels in the system. I review how much petrol and diesel are currently available in the depot and compare them with the previous day's closing balance. If fuel stock is getting low, I inform management so they can arrange a new fuel delivery.

When a company vehicle comes for refuelling, I first identify the vehicle using its registration number. I check the vehicle record in the system and verify that it is authorised to receive fuel.

After the fuel is pumped, I create a fuel-log and I enter the vehicle number, fuel type, quantity issued, current odometer reading, date, time, and fuel cost. Once saved, the system

Kasun here again, are you preparing any reports daily. if so can you provide me an example

Project Discussion Bot - Year2

Kasun Senanayake – Fuel Station Manager (Colombo Depot)

Hello Amantha. Yes, I prepare a **Daily Fuel Usage Report** at the end of every day. This report helps me monitor fuel issues, vehicle consumption, and remaining stock.

Here is a simple example:

Daily Fuel Usage Report

Date: 24/07/2026

Vehicle No	Fuel Type	Quantity (L)	Odometer (km)	Cost (LKR)	Driver
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SAMPLE 06

Niroshan / Validation · roles, security, audit, capacity and retention

hey Niroshan,Im Afsal. I will be attending to gather requirements for the system you want, please explain what needs to be done. what are the day to day tasks you do?

Project Discussion Bot - Year2

Niroshan Perera – Workshop Operations Manager

Hello Afsal. Thank you for gathering requirements for us. My main responsibility is making sure all vehicle services are completed on time and that the workshop operates smoothly.

A typical day starts with me checking all service appointments scheduled for the day. I review the vehicles coming in, the type of service required, available technicians, and available service bays. Based on the workload, I plan how work should be assigned.

When a vehicle arrives, I verify the booking details and assign the vehicle to a suitable service bay. Then I assign a technician who has the right skills for that service

do you currently have a system for this? if so let me know I need to observe

Project Discussion Bot - Year2

Niroshan Perera – Workshop Operations Manager

Hello Afsal.

Yes, we currently use a basic system, but it is quite limited. It allows us to record vehicle service bookings, update service status, and maintain some vehicle service history. However, many activities still require phone calls, spreadsheets, or manual coordination between departments.

If you were observing my work, I would show you:

1. Daily Service Schedule Screen

List of bookings for the day

A Stakeholders and Access Model

Every named interview voice, wider stakeholder group and expected permission boundary

SECONDARY STAKEHOLDER LANDSCAPE

Primary
Customers, registered users, administrators, scheduling and booking officers, station/depot managers, technicians, fuel attendants and customer service.

Management
Operations, regional, finance, revenue, general and executive management.

Technical
System and database administrators, IT support, software maintenance and cybersecurity teams.

Financial
Payment gateways, banks, accounting teams and auditors.

Regulatory
Ministry of Transport, fuel regulatory authorities, government auditors and regulators.

External
Fuel suppliers, insurance providers, spare-part vendors and corporate fleet clients.

Operational
Bay technicians, dispatch officers, booking clerks, control-room staff and maintenance teams.

Boundary
These groups support or use the system. They are not extra core CRUD allocations for the six members.

PERSON / ROLE	DETAILED RESPONSIBILITY AND PERMITTED SYSTEM AREA	DECISION OR PRIVACY BOUNDARY
Niroshan Perera Workshop Operations Manager	Monitor bay/technician availability and operations; manage delays, breakdowns and incidents; review operational reports; approve booking cancellations.	Senior Operations Officers may cancel; General Manager handles major situations. The exact major-incident rule is open.
Dilmi Jayasinghe Customer Service	View bookings; answer booking questions; register complaints; create refund requests; track customer cases; send customer communication.	Can request a refund but Finance approves it. Access only the customer data needed for service.
Ruwan Fernando Service Scheduling Officer	Create and update service schedules; allocate bays/technicians; view demand and occupancy; analyse capacity; recommend cancellations and schedule changes.	Normally receives aggregate statistics, not bank, complaint or refund details. Personal access needs an authorised investigation.
Malsha Wickramaratne Finance & Revenue	View payments; reconcile gateway and bank records; track revenue and duplicate transactions; approve refunds; use financial reports.	Finance Manager and authorised Finance Officers approve. Automatic refund limits still need approval.
Kasun Senanayake Fuel Station Manager	Monitor fuel dispensing; assign bays; publish vehicle information; supervise fuel discrepancies; record incidents; review daily station reports.	Station staff see operational data for their duties. Sensitive finance and investigation data stays restricted.
Public / guest	View services, fuel plans, prices, station locations, announcements, contact details, FAQs and terms. Guest checkout is proposed.	No internal operational or financial records. Secure guest invoice/refund retrieval is open.
Registered customer	Own bookings, invoices, history, refunds, complaints, saved vehicles, profile, personalised notices and fuel logs.	May view only their own records.
Administrator / management	Admin: users, roles, configuration and audit logs. Management: read-only strategic dashboards, revenue and performance reports.	"Full access" boundaries and field-level permissions require final approval.

Confidentiality rule: Actual company reports contain confidential business information. Report examples in this document are illustrative requirements only, and report access must follow role permissions.

B

Workshop and Fuel Station Operations

Daily station work, demand planning, breakdown impact and continuity needs

KASUN'S CONFIRMED DAILY STATION WORKFLOW

01

Review the day

Check timetable, bookings, maintenance and special-service notices.

02

Monitor vehicles

Track arrivals, departures and whether services follow the schedule.

03

Confirm delays

Contact Operations and nearby stations to confirm the cause and current situation.

04

Manage bays

Assign before arrival and reassign after changes to avoid conflicts.

05

Inform customers

Provide current service times, bays, delays and cancellations.

06

Support booking

Handle escalated reservation, bay-availability and customer issues.

07

Record incidents

Capture operational, breakdown, fuel and security details and route them for action.

08

Close the day

Review movements, delays, incidents, customer issues and operational notes.

Current Working Sources

- Internal systems, station registers, spreadsheets, reports, phone calls and official messages. Information arrives from several places, so staff need time to form one current view.

Proposed Live Station Dashboard

One screen for current bays, delays, fuel stock, customer announcements and open incidents. Refresh rate, source priority and offline fallback remain open.

Movements

Next arrivals and departures

Bays

Assignments and conflicts

Fuel Stock

Levels and deliveries

Announcements

Latest customer message

Incidents

Open issue and owner

RUWAN'S CONFIRMED PLANNING AND CHANGE-IMPACT NEEDS

Demand and Capacity

- Bookings per service type and center; date, class and bay-occupancy status.
- Compare bay capacity with booked slots.
- Recommend extra bays, more technicians, or timetable adjustments.
- Compare public holidays, weekends, weekdays and seasonal patterns.

Schedule Change Impact

- Booking number, center and old/new appointment time.
- Affected customer count and authorised contact list.
- Alternative bay/technician availability.
- Reassign where possible, or provide alternative/refund and send notices.
- Scheduling recommends; authorised Operations decides cancellation.

INTEGRATIONS AND OPERATING CONTINUITY

EXPECTED

Payment gateway/banks, SMS/email, existing workshop system, fuel-tracking system if retained, and operations-data systems.

FUTURE

Mobile application and government verification services.

OPEN

Procedures for gateway outage, full network failure or one station losing connectivity.

C

Data Dictionary and Special Records

Fields named by stakeholders, conditional data and least-access rules

PROCESS / ENTITY	FIELDS REQUESTED IN THE CONVERSATIONS	STATUS	ACCESS NOTE
User account	Full name, NIC/license no., date of birth, mobile, email, username and protected password.	PROPOSED	Registration set; required/optional rules need approval.
Vehicle	Registration no., make/model, fuel type, engine capacity, owner and last service/mileage.	ELICITED	Owner and technician access only.
Booking	Full name, NIC, mobile, optional email, vehicle, service type, center, date, technician preference.	ELICITED	Preference applies only where offered.
Cancellation	Booking number, customer name, contact number, cancellation reason and request date.	ELICITED	Final deadline and fee rules remain open.
Refund	Booking number, customer details, payment reference, reason, method, conditional bank details, amount and status.	ELICITED	Protect bank/refund data; approval limits remain open.
Complaint	Complaint ID, customer/contact, category, description, date/time, related booking, evidence, assigned owner and status.	ELICITED	Customer Service and assigned department only.
Payment	Booking ID, transaction ID, method, date/time, amount paid, payment status, refund amount and refund status.	ELICITED	Keep gateway references; do not store raw card credentials.
Fuel log	Vehicle, station, fuel type, quantity, odometer reading, amount and date/time.	ELICITED	Station staff access by duty.
Notification lookup	Customer name, contact number, email, booking reference and vehicle number for affected-customer contact.	ELICITED	Release contact lists only for authorised service, breakdown or emergency work.

PROPOSED DATA ENTITIES

Customer

User Account

Booking

Invoice

Payment

Refund

Complaint

Vehicle

Fuel Log

Notification

Emergency Contact

EMERGENCY AND INCIDENT RECORD FIELDS

Vehicle Breakdown

Vehicle, driver, location, time and action taken.

Fuel Spillage

Incident ID, date, location, cause and response actions.

Fire

Location, emergency actions, injuries and response team.

Security Threat

Threat type, reporter, location and investigation information.

Accident / Collision

Affected vehicles, delay information and customer notifications.

Fuel Contamination

Batch details, affected vehicles, search information and resolution details.

Conditions still open: fallback when email is unavailable; when technician preference applies; when bank details or an emergency contact are necessary; and who authorises Scheduling to see personal data during a special investigation.

D Report and Analytics Catalogue

Illustrative fields and measures requested by each department

Evidence note: The stakeholder did not share real company reports because they are confidential. The schemas below are examples of what the new system should support, not approved report templates.

REPORT	DETAIL COLUMNS	SUMMARY / MEASURE	MAIN USER
Daily Revenue	Report date, center, services sold and route revenue.	Total revenue, highest and lowest center, refund count.	Malsha / management
Refund	Refund ID, booking, customer, reason, amount and status.	Total requests, approved, rejected and total refunded amount.	Finance / service
Payment Reconciliation	Payment method, booking amount, bank amount and difference.	Successful, failed, unmatched and reconciliation status.	Malsha / Finance
Daily Service Operations	Vehicle, center, planned time and status.	Completed, on-time, delayed and cancelled totals.	Niroshan / management
Delay	Vehicle, center, delay duration and reason.	Delay count, average duration and most common reason.	Operations / station
Schedule Change	Date, booking, old schedule, new schedule and reason.	Total changes, delays, and temporary cancellations.	Ruwan / Operations
Station Operations (Fuel)	Vehicle, bay, arrival, departure and status.	Bay use, customer incidents, disruptions and notes.	Kasun / station
Customer Complaint	Complaint ID, category, status and assigned department.	Total, open, closed and average resolution time.	Dilmi / management
Occupancy / Demand	Center, capacity, booked slots and occupancy percentage.	Demand by center/date, peak hour, and schedule-change impact.	Ruwan / management

NAMED REPORT FAMILIES

- RevenueService SalesRefundPaymentVehicle OperationsDelayCancellationScheduleStationCustomerComplaint
- Fuel UtilisationRoute PerformanceCustomer DemandPeak-HourSchedule Change ImpactFinancial SummaryDashboard Summary

REPORT DISTRIBUTION AND CONTROLS

ROLE BASED

Management: strategic revenue, demand, utilisation, complaints, refund and performance dashboards.

DEPARTMENT

Service: complaints/booking/refund status. Station: bay/movement. Scheduling: occupancy/demand. Finance: payment/refund/revenue.

OUTPUT

Filter by invoice, vehicle, customer, NIC, booking, date range and center. Export PDF, Excel or CSV if approved.

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Additional CRUDs and Supporting Rules

Important supporting areas - not additional team-member allocations

Maintenance Schedule/history/service record, availability, notices and due reminders. Unavailable vehicle alerts Scheduling.	Bay and Technician Bay details/type/allocation, technician roster and skill categories.	Pricing / Fare Create service/fuel price list; seasonal prices, discounts and price history.	Passenger/Customer Security Flag Fraud, payment abuse or security flag; investigation record and restricted-booking control.
Group Booking Fleet, corporate or family vehicles under one booking; group management. Discount rules are open.	Waiting List Bays full → add customer → bay released → notify and offer slot.	Feedback Service completed → request → 1-5 rating, comment, center/date → management review.	Lost and Found Record lost item, found item, claimant and claim outcome.
Document Repository Operational notices and incident, maintenance and investigation reports.	Special Services Temporary campaign, seasonal check-up and emergency roadside schedules.	Loyalty - Future Membership, frequent-customer points, benefits and approved discounts.	Employee Metrics Complaints resolved, refund processing time and schedule-change handling time.

EXCEPTION DETAILS THAT MUST STAY IN SCOPE

Payment and Invoice - Failure causes: gateway error, network interruption or insufficient funds. - Payment success + invoice failure: incident, automatic retry, staff review and recoverable booking. - Duplicate transaction ID: flag for Finance and refund extra payment when confirmed. - Normal late confirmation: Pending Verification, then issue invoice. After a released bay remains open. - Booking confirmation contains booking ID, invoice number and service information.	Customer and Service - No bay: suggest another bay, another center or waitlist. - Lost invoice: secure reissue/retrieval. No-show rule remains open. - Bay/technician/schedule change: identify customers, reassign or offer alternative/refund, then notify. - Account recovery uses password reset and proposed OTP verification. - Fleet reservation restrictions, invoice after service and final refund deadline need approval.
ARCHIVE PROPOSAL Archive completed bookings, historical schedules, old complaints/refunds, operational/revenue/incident reports, audit logs and service history. Final deletion and anonymisation follow approved policy and law.	NOTIFICATION PROPOSAL Delay: service end. Cancellation: 30 days. Refund: completion +30 days. Booking confirmation: booking history. Emergency: until cleared. Final policy remains subject to approval.
Final unresolved choices: multi-vehicle bookings; final bay-hold and expired-hold payment action; refund percentages and amount limits; secure guest retrieval; offline duplicate invoice handling; exact admin/field permissions; peak load; browser/device/accessibility targets; vendor APIs; report frequency/retention; and final deletion rules.	

CONCLUSION AutoFlow will provide one simple platform for vehicle service booking, fuel tracking, payments and customer updates. The gathered requirements will help the team build a reliable system that reduces manual work and makes vehicle care easier for both customers and workshop staff.
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